

Jessica Pieraccini Kunke

Appointments

2025–present **Assistant Professor of Statistics**, *Department of Mathematical Sciences, Montana State University*, Bozeman MT.

Education

- 2025 **Ph.D. in Statistics**, *University of Washington*, Seattle WA.
Thesis: Leveraging network information to improve population size estimation in social and environmental applications, advised by Tyler H. McCormick
- 2020 **M.S. in Statistics**, *University of Chicago*, Chicago IL.
Thesis: A Nonstationary Gaussian Process Model for Forecasting Solar Irradiance at Earth's Surface, advised by Mihai Anitescu and Peter McCullagh
- 2016 **A.M. in Earth and Planetary Science**, *Harvard University*, Cambridge MA.
- 2010 **B.A. in Integrated Science, Earth and Planetary Science, Mathematics**, *Northwestern University*, Evanston IL.

Publications

Preprints

- 2025 **Kunke, J. P.**, Olden, J. D., and McCormick, T. H. (2025), "Scalable Spatial Stream Network (S3N) Models," *arXiv preprint arXiv:2512.12398*.

Published Papers

- 2025 **Kunke, J.** (2025), "Leveraging network information to improve population size estimation in social and environmental applications," Ph.D. thesis, University of Washington.
- 2025 Banks, D., **Kunke, J.**, and Zejnullahi, R. (2025), "How Can Statisticians Support Refugees?" *CHANCE*, 38, 6–7.
- 2025 Laga, I., **Kunke, J. P.**, McCormick, T. H., and Niu, X. (2025), "Estimating and Correcting Degree Ratio Bias in the Network Scale-up Method," *Sociological Methods & Research*, 00491241251364233.
- 2024 **Kunke, J. P.**, Laga, I., Niu, X., and McCormick, T. H. (2024), "Comparing the Robustness of Simple Network Scale-Up Method Estimators," *Sociological Methodology*, 54, 385–403.
- 2023 **Kunke, J. P.**, Visokay, A., and McCormick, T. H. (2023), "Respondent-driven sampling: An overview in the context of human trafficking," *CHANCE*, 36, 15–21.
- 2021 Wang, Z., Peterson, C., Zhou, Q., Subramanian, R., **Kunke, J.**, Baker, A., Moyer, E., and Hammerling, D. (2021), "Characterizing Internal Variability and Detecting Changes in Model and Computational Parameters in a Century-Long CESM Ensemble," NCAR Tech. Note NCAR/TN-569+STR, National Center for Atmospheric Research.

Software

- 2025 **S3N**, <https://github.com/jpierkunke/S3N>.
- 2025 **dataQCtools**, <https://github.com/tribalxg/dataQCtools>.

Grants

- 2025 **USDA**, *Co-PI*, National Institute of Food and Agriculture (NIFA) New Beginning for Tribal Students, Award 2021-70411-35304 *Returning to Our Good Camp*, PI: Margaret Eggers, Co-PIs: Vanessa Simonds, Susan Kelly, John T. Doyle, 09/01/2021-08/31/2026 (joined post-award as Co-I 11/2025, Co-PI 02/2026), \$1,123,425.
- 2025 **MSU College of Letters and Science**, *PI*, Research Catalyst Grant, 10/17/2025-10/16/2026, \$5,000.

Awards and Honors

- 2024 **American Statistical Association GSS/SRMS/SSS Student Paper Competition**.
- 2023 **University of Washington Excellence in Teaching Award Finalist**.
- 2022 **Dorothy M. Gilford Teaching Award** for outstanding performance by a graduate teaching assistant.
- 2020 **Achievement Rewards for College Scientists (ARCS) Fellowship (three years)**.
- 2019 **Senior Consultant Award**, *Department of Statistics Consulting Program, University of Chicago*.
- 2018 **NSF Research Traineeship (NRT) Grant Trainee (two years)**, *PI Elisabeth Moyer*.
- 2018 **Senior Consultant Award**, *Department of Statistics Consulting Program, University of Chicago*.

Teaching

Montana State University

- Fall 2026 **STAT 446 Sampling**, undergraduate course, XX students, 3 credits.
- Fall 2026 **STAT 436 Time Series Analysis**, undergraduate course, XX students, 3 credits.
- Spring 2026 **STAT 408 Statistical Computing**, undergraduate course, 34 students, 3 credits.
- Fall 2025 **STAT 446 Sampling**, undergraduate course, 30 students, 3 credits.
- Fall 2025 **STAT 421 Probability Theory**, undergraduate course, 30 students, 3 credits.

University of Washington (Instructor of record as doctoral student)

- Fall 2024 **CSSS 508 Intro to R for the Social Sciences**, graduate course, 1 credit.
- Summer 2024 **STAT 302 Statistical Computing**, undergraduate course, 3 credits.
- Summer 2022 **STAT 394 Probability Theory I**, undergraduate course, 3 credits.

Workshops, Tutorials and Demos

- May 2026 **Tribal Water Quality in R (demo)**, *EPA Region 8 Water Quality Training*, Tribal environmental professionals, virtual.
- Mar 2026 **Intro to GitHub, Quarto dashboards, and dataQCtools**, *Pacific Northwest Tribal Coding Group in-person meeting*, Tribal environmental professionals, Tacoma WA.
- Mar 2026 **Intro to Data Entry and Analysis with Excel**, *Alaska Tribal Conference on Environmental Management (ATCEM)*, Tribal environmental professionals, Anchorage AK.
- Feb 2026 **High performance computing with Tempest (tutorial)**, *MSU Department of Mathematical Sciences*, students and faculty, Bozeman MT.
- Nov 2025 **Advanced R Workshop**, *Tribal Exchange Network*, Tribal env. professionals, Tulalip WA. National Association of Wetland Managers (NAWM) Pacific NW Tribal Clean Water Act Training
- Mar 2025 **Collaborating via GitHub (demo)**, *Pacific Northwest Tribal Coding Group in-person meeting*, Tribal environmental professionals, Tacoma WA.
- Sep 2025 **Gaussian Processes (tutorial)**, *MSU TAILS seminar*, students and faculty, Bozeman MT.
- Sep 2024 **CSSS Math Camp**, *University of Washington*, graduate students, hybrid (Seattle WA).
- Sep 2023 **CSSS Math Camp**, *University of Washington*, graduate students, hybrid (Seattle WA).

- Jul 2023 **R Workshop**, *Tribal Exchange Network*, Tribal environmental professionals, virtual.
- Sep 2022 **CSSS Math Camp**, *University of Washington*, graduate students, virtual.
- Nov 2021 **R for Ecologists**, *Tribal Exchange Network*, Tribal environmental professionals, virtual.
- Nov 2020 **R for Ecologists**, *Data Carpentries + Tribal Exchange Network*, Tribal env. pro., virtual.
- Sep 2019 **Statistics and R Bootcamp**, *University of Chicago*, students and researchers in other disciplines, Chicago IL.

Academic Service

Departmental, Mathematical Sciences, Montana State University

- Apr 2026 **DataFest Judge**.
- AY 25-26 **Actuarial Advisor**.
- AY 25-26 **Graduate Admissions Committee**.
- AY 25-26 **Alumni Panel Co-Organizer**.
- AY 25-26 **Graduate Admissions Committee**.

Professional

- Jun 2026 **WNAR Student Paper Competition Judge**.
- Sep 2025 **Guest Editor**, *CHANCE special issue, Volume 38, Issue 4*, Refugee Crisis: How Statisticians Can Help.
- 2025- **Peer Reviewer**, *Journal of Open Source Education (JOSE)*, *Journal of Statistics and Data Science Education (JSDSE)*, *Public Library of Science (PLOS) One*, *Social Networks*, *Social Methodology*.

Presentations

Invited Talks

- Jun 2026 **WNAR**, Pullman, WA.
Scalable spatial stream network models
- Mar 2026 **USGS Lunch and Learn**, virtual.
Scalable spatial stream network models
- Jan 2026 **IAD Seminar**, virtual.
Scalable spatial stream network models

Additional Presentations

- Aug 2026 **JSM 2026**, *Talk*, Boston, MA.
Scalable spatial stream network models
- Oct 2026 **ASA Montana Chapter Meeting**, *Talk*, Bozeman, MT.
Scalable spatial stream network models
- Oct 2024 **Tribal Exchange Network Conference (TENC)**, *Roundtable discussion*, Remote.
Tribes bRw coding collaborations
- Oct 2024 **ENVR 2024**, *Poster*, Boulder, CO.
Scalable spatial stream network models
- Aug 2024 **JSM 2024**, *Student award presentation*, Portland, OR.
Comparing the robustness of simple network scale-up method (NSUM) estimators
- Jun 2024 **eCOTS 2023**, *Poster*, virtual.
Resources for JEDI-informed statistics teaching
- Oct 2023 **WSDS 2023**, *Panel proposer and chair*, Bellevue, WA.
Frontiers, challenges and progress in human trafficking-related statistics

Aug 2023 **JSM 2023**, *Poster*, Toronto, Canada.

Network scale-up model (NSUM) estimators: Is the average of ratios or ratio of averages more robust?